**EL MOUATAMID Abdellatif**
*Junior Professor Chair, Espace-Dev Research Unit*, University of French-Guiana
*Ph.D. in Energy Engineering and Automation*

Email: abdellatif.elmouatamid@gmail.com
Phone: +594 694249527
WhatsApp +212 649-966652
Address: Sciences and Technology Department, UMR Espace-Dev Lab, Energy Transition Team, University of French Guiana, Cayenne, France.
Homepage https://elmouatamidabdellatif.github.io/github.ma/

## Biography

Elmouatamid Abdellatif is a Junior Professor Chair at the University of French Guiana, France. He was a Postdoctoral research associate at the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, USA (2022-2024). He was in a postdoctoral position at IRESEN (Institute of Research in Solar Energy and New Energies), Morocco (2021-2022). He received a Ph.D. in Energy Engineering and Automation from Chouaib Doukkali University of El-Jadida, Morocco (Dec-2020). He received a Master from Chouaib Doukkali University with 1st class honors (2016). He has research experience in transmission and distribution lines, power systems, energy efficiency in buildings, smart microgrids, energy management, anomaly detection, and predictive maintenance, and engaged in numerous collaborative research projects.

## Professional Experience

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| 2024-Present | Junior Professor Chair at the Department of Sciences and Technology, University of French-Guiana, Espace-Dev Research Unit, France. |
| 2022-2024 | Post-Doctoral Research Associate at the Department of Electrical and Computer Engineering, New Jersey Institute of Technology, United States. |
| 2021-2022 | Post-Doctoral Researcher, IRESEN (IRESEN-Solar Energy and New Energies Research Institute), Morocco. |

## Education

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| 2017-2020 | Ph.D. in Energy Engineering and Automation, Chouaib Doukkali University, Faculty of Sciences, Morocco (December 2020). |
| 2014-2016 | Master's degree in industrial Electronics & Telecommunications, Chouaib Doukkali University, Faculty of Sciences, Morocco. |
| 2010-2014 | Bachelor's degree in information and communication technologies, Chouaib Doukkali University, Faculty of Sciences. |
| 2010 | Baccalaureate diploma in sciences. |

## Editorial Activities and Committees

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| 2023-Present | Guest editor (Special Collection): Fault Detection, Classification, and Diagnostic in Microgrid Systems Infrastructure Using Data Mining Approaches, Journal of Measurement and Control. |
| 2022-Present | Editorial Board Membership of International Journal of Sensors and Sensor Networks (IJSSN) ISSN Print: 2329-1796; ISSN Online: 2329-1788. |
| 2022-Present | Associate member in STIC-Lab (Technological Systems of Information and Communication Laboratory), Chouaib Doukkali University, Faculty of Sciences, Morocco. |
| 2021-Present | Volunteer Reviewer Board of Multidisciplinary Digital Publishing Institute (MDPI) journals: Energies, Electronics, Sensors, Smart Cities. |
| 2020-Present | Volunteer Reviewer Board of IEEE Access Journal. |

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| 2023 | Guest editor (Special Issue): Latest Developments in Offshore Wind Technologies, Energies Journal. |
| 2023 | Bootcamp Organizer: New Jersey 2-Day Offshore Wind Bootcamp Training, New Jersey Institute of Technology, USA. |
| 2023 | Community member: GREEN-2022 and GREEN-2023, The Eighth International Conference on Green Communications, Computing and Technologies, Portugal. |
| 2022 | Speaker/Organizer: North Jersey Research Student Conference 2022, New Jersey Institute of Technology, USA. |
| 2020 | Technical Program Committee Member, The International Workshop on Emerging Networks and Communications (IWENC 2019), Nov 2020, Coimbra, Portugal. |
| 2019 | Community members of the 7th Job-Fair event at the International University of Rabat hosted many stands for more than 35 companies from different fields. |

## Relevant Training & Experiences

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| Mar 2023 | OPAL-RT: Hardware-in-the-Loop and Real-Time Simulation Training. |
| Dec 2019 | Internship: UMI-GT-CNRS (*Unité Mixte Internationale-Georgia Technologie-Centre National de la Recherche Scientifique*) in Georgia Tech Lorraine, France.<br>Topic: The development of photovoltaic energy research to establish protocols, and tests in common between GT-CNRS and the UIR (International University of Rabat). |
| Nov 2019 | Workshop trainer: IoT/Big-Data technologies for embedded systems, ENSA (National School of Applied Sciences) Marrakech. |
| Aug-Sept 2019 | Participant in the international competition Solar Decathlon AFRICA: Renewable energy sources and data monitoring platforms installation, held in Green Energy Park, Benguerir, Morocco. |
| Dec 2018 | TRNSYS Certificate. |
| Oct 2018 | Raspberry-Pi, Arduino Developer Certificate. |
| Oct 2018 | Patent writing Certificate, OMPIC (Office Marocain de la Propriété Industrielle et Commerciale). |
| Mar 2018 | Training workshop: Patent conscripting and drafting, OMPIC. |
| Jul 2017 | Training: Course LabVIEW, Core 1 and Core 2. |
| Apr-Jun 2016 | Internship: Study of Industrial Sensors for surveillance, within the College of Sciences in El-Jadida, STIC laboratory (*Systèmes Technologiques de l'information et de communication*). |
| May 2015 | Microsoft Certified Solutions Associate: Windows Server 2012, Server Infrastructure. |
| Mar 2013 | Certificate in Cisco Certified Network Associate (CCNA). |
| Aug 2013 | Initiation Internship within OCP (Office Chérifien des Phosphates) Group in Jorf-Lasfar:<br>Study of the existing Network in the region of Jorf-Lasfar and Migration to IPV6 using GNS3, VMWARE WORKSTATION, PAQUET TRACER, and Wireshark. |

## Synergistic Activities

**THyGRE**
2024-2029
ANR-UG

- Project Manager: Hydrogen Transition Applied to Electrical Grid Management in Non-Interconnected Areas.

The research project aims to develop expertise on the site based on two objectives: (i) to develop innovative methods for managing multi-source systems; (ii) to characterize the services that hydrogen storage can provide to off-grid networks.

**HOLSYS**
2019-2021
UIR-IRESEN

- Postdoc member of HOLSYS Project: UIR-IRESEN-ENSAK.

The project developed a Software/Hardware platform for real-time data collection and communication interfaces in Microgrid systems for both energy and comfort management.

**MIGRID**
2017-2020
UIR- UCD

- Ph-D Student member of MIGRID Project: USAID-PEER Project.

The project developed new approaches for Renewable Energy Sources integration in the buildings with fuel cells and battery storage system Energy Efficiency.

**Solar Decathlon AFRICA**
Aug-Sep 2019
UIR-IRESEN

- Ph-D Student member of Solar Decathlon AFRICA Project, IRESEN, Morocco.

The project challenges collegiate teams to design and build houses powered exclusively by the sun, to increase energy efficiency, and to construct buildings with local materials.

## Publications

***Journals Papers:***


[1] **Elmouatamid**, A., Chowdhury, M. A., Netto, M., & Pong, P. W. (2024). ''Economic feasibility study of standalone community microgrid in 37 cities of the USA'', HKIE Transactions, Volume 31 - Issue 1, p. 1-17.

[2] **Elmouatamid**, A., Brian F., Jian S., and Philip W. T. P. (2023). "Air Conditioning Systems Fault Detection and Diagnosis-Based Sensing and Data-Driven Approaches" Energies 16, no. 12: 4721.

[3] **Elmouatamid**, A., Ouladsine, R., Bakhouya, M., El Kamoun, N., & Zine-Dine, K. (2021). "*A Predictive Control Strategy for Energy Management in Micro-Grid Systems"*. Electronics.

[4] **Elmouatamid**, A., Ouladsine, R., Bakhouya, M., El Kamoun, N., Zine-Dine, K., & Khaidar, M. (2020). "*Review of Control and Energy Management Approaches in Micro-Grid Systems"*. In Smart Grids and Microgrids. Energies.

[5] **Elmouatamid**, A. (2020). MAPCAST: "A*n Adaptive Control Approach using Predictive Analytics for Energy Balance in Micro-Grid Systems"*. International Journal of Renewable Energy Research (IJRER), 10(2), 945-954.

[6] **Elmouatamid, A**., NaitMalek, Y., Bakhouya, M., Ouladsine, R., Elkamoun, N., Zine-Dine, K., & Khaidar, M. (2019). "*An energy management platform for micro-grid systems using Internet of Things and Big-data technologies"*. Proceedings of the Institution of Mechanical Engineers, Part I: Journal of Systems and Control Engineering, 0959651819856251.

[7] Raj, R., **Elmouatamid**, A., Swain, A., & Pong, P. W. (2025). Critical review of hybrid modular multilevel converters for HVDC transmission in offshore wind energy. Energy Reports, 14, 1761-1778.

[8] Elkhoukhi, H., **Elmouatamid**, A., Haibi, A., Bakhouya, M., & El Ouadghiri, D. (2025). Overview of Sensing and Data Processing Technologies for Smart Building Services and Applications. Sustainability, 17(9), 4029.

[9] Elkhoukhi, H., Bakhouya, M., **Elmouatamid**, A., El Ouadghiri, D. and Gaber, J., 2025. A review of occupancy sensing technologies and approaches in smart buildings. *International Journal of RF Technologies*, *15*(1), pp.19-48.

[10] Akhyurna, S., **Elmouatamid**, A., Ahmed, M., Philip W.T. (2022). "*Sensor Technologies for Transmission and Distribution Systems: A Review of the Latest Developments. In Condition Monitoring of Critical Infrastructure for Energy Systems"*, Energies-1917449.

[11] Elouadoud, H., Ouladsine, R., **Elmouatamid**, A., Bakhouya, M., & Pierfederici, S. (2022). "*Towards an Advanced Control Approach for Energy Management in Distributed Micro-Grid Systems"*. Available at SSRN 4191086.

[12] Alidrissi, Y., Ouladsine, R., **Elmouatamid**, A., Errouissi, R., & Bakhouya, M. (2022). "*Constant Power Load Stabilization in DC Microgrids Using Continuous-Time Model Predictive Control".* Electronics, 11(9), 1481.

[13] Alidrissi, Y., Ouladsine, R., **Elmouatamid**, A., & Bakhouya, M. (2021). "*An Energy Management Strategy for DC Microgrids with PV/Battery Systems"*. Journal of Electrical Engineering & Technology, 1-12.

[14] S. Boulmrharj, Y. NaitMalek, **A. Elmouatamid**, M. Bakhouta, R. Ouladsine, K. Zine-dine, M. Khaidar, M. Siniti. "*A Dimensioning Approach for Micro-Grid Systems"*. Energies Journal 2018.


***Chapter Book:***


[15] **Elmouatamid A.** et al. (2021) "*Towards an IoT/Big-Data Platform for Data Measurements, Collection and Processing in Micro-grid Systems"*. In: Motahhir S., Bossoufi B. (eds) Digital Technologies and Applications. ICDTA 2021. Lecture Notes in Networks and Systems, vol 211. Springer, Cham. https://doi.org/10.1007/978-3-030-73882-2_26

[16] **Elmouatamid**, Abdellatif, et al. "*A microgrid system infrastructure implementing IoT/Big-Data technologies for efficient energy management in buildings*." Advanced Technologies for Solar Photovoltaics Energy Systems. Cham: Springer International Publishing, 2021. 571-600.

***Patents:***

[17] **A. Elmouatamid,** Y. NaitMalek, S. Boulmrharj, M. Bakhouya, R. Ouladsine, M. El kamoun, K. Zine-dine, M. Khaidar "*Procédé de monitoring, estimation, prédiction et contrôle en temps réel des systèmes multi-sources*", National publication No: MA 43611 A1, International CI: G01R 21/00; G01R 21/00

***Conferences:***

[18] Erik, L., **Elmouatamid**, A., & Pong, P. W. (2025, December). Anomalous Conditions Analysis of HVDC and Multi-Terminal HVDC Systems for Wind Farm Integration. In *2025 New Jersey Future Energy Transmission Conference (NJFET)* (pp. 1-6). IEEE.

[19] Kabir, R., **Elmouatamid**, A., Elkhoukhi, H., & Pong, P. W. (2024, February). Photovoltaic Power Forecasting Using Neural Networks for Short and Medium-Term Dependencies. In 2024 IEEE Texas Power and Energy Conference (TPEC) (pp. 1-6). IEEE.

[20] Swain, Akhyurna, **Elmouatamid** Abdellatif, and Philip WT Pong. "Magnetic Signature-Based Model Using Machine Learning for Electrical and Mechanical Faults Classification of Wind Turbine Drive Trains." 2024 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT). IEEE, 2024.

[21] **Elmouatamid**, A., Ouladsine, R., Bakhouya, M., Zine-Dine, K., & Khaidar, M. (2019, October). A control strategy based on power forecasting for micro-grid systems. In *2019 IEEE International Smart Cities Conference (ISC2)* (pp. 735-740). IEEE.

[22] **Elmouatamid**, A., Ouladsine, R., Bakhouya, M., El Kamoun, N., Zine-Dine, K., & Khaidar, M. (2019, September). A model predictive control approach for energy management in micro-grid systems. In *2019 international conference on smart energy systems and technologies (SEST)* (pp. 1-6). IEEE.

[23] **Elmouatamid**, A., Bakhouya, M., Ouladsine, R., Zine-Dine, K., Khaidar, M., & Abid, R. (2018, December). Deployment and experimental evaluation of micro-grid systems. In *2018 6th international renewable and sustainable energy conference (IRSEC)* (pp. 1-6). IEEE.

[24] **Elmouatamid**, A., NaitMalek, Y., Ouladsine, R., Bakhouya, M., Elkamoun, N., Zine-Dine, K., ... & Abid, R. (2018, April). Towards a Demand/Response Control Approach for Micro-grid Systems. In *2018 5th International Conference on Control, Decision and Information Technologies (CoDIT)* (pp. 984-988). IEEE.

[25] **El Mouatamid**, A., Ouladsine, R., Bakhouya, M., Felix, V., Elkamoun, N., Zine-Dine, K., ... & Abid, R. (2017, December). Modeling and performance evaluation of photovoltaic systems. In *2017 International renewable and sustainable energy conference (IRSEC)* (pp. 1-7). IEEE.

[26] **Elmouatamid**, A., Najib, E., Fatima, L., & Mohamed, B. (2018, May). Wireless sensor networks for smart micro-grids. In *Conference*.

[27] Elibrahimi, M., **Elmouatamid**, A., Bakhouya, M., Feddi, K., & Ouladsine, R. (2018, December). Performance evaluation of fixed and sun tracking photovoltaic systems. In *2018 6th International Renewable and Sustainable Energy Conference (IRSEC)* (pp. 1-6). IEEE.

[28] Bakhouya, M., NaitMalek, Y., **Elmouatamid**, A., Lachhab, F., Berouine, A., Boulmrharj, S., ... & Elkamoune, N. (2017, October). Towards a context-driven platform using IoT and big data technologies for energy efficient buildings. In *2017 3rd international conference of cloud computing technologies and applications (CloudTech)* (pp. 1-5). IEEE.

[29] Boulmrharj, S., NaitMalek, Y., **Elmouatamid**, A., Bakhouya, M., Ouladsine, R., Zine-Dine, K., ... & Abid, R. (2018). Approach for dimensioning stand-alone photovoltaic systems. *Energy Procedia*, *153*, 56-61.

[30] Boulmrharj, S., NaitMalek, Y., **El Mouatamid**, A., Ouladsine, R., Bakhouya, M., Ouldmoussa, M., ... & Abid, R. (2018, September). Towards a battery characterization methodology for performance evaluation of micro-grid systems. In *2018 International Conference on Smart Energy Systems and Technologies (SEST)* (pp. 1-6). IEEE.

## Teachings

| Years - Level | | Institution | Course | Hours |
|---|---|---|---|---|
| 2024-2026 | Master | University of French Guiana<br>Master: Energy | Energy sources and storage systems | 42×2 |
| 2024-2026 | Master | University of French Guiana<br>Master: Energy | Artificial intelligence for power systems | 42×2 |
| 2024-2026 | Master | University of French Guiana<br>Master: Energy | Energy conversion | 42×2 |
| 2024-2026 | Master | University of French Guiana<br>Master: Energy | Electrical Machines | 42×2 |
| 2024-2026 | Master | University of French Guiana<br>Master: Energy | Research/innovation projects supervising | 42×2 |
| 2023-2024 | Master | ECE 449 - NJIT | Energy management | 24 |
| | | | Electricity transport and distribution systems | 24 |

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| | | Master: Electrical system and Energy production | | |
| 2021-2022 | Engineering school | ECINE 3-S5<br>International University of Rabat | Control Signals | 20 |
| | | | Electronic Signals and Systems: Signal Processing and Conception | 20 |
| 2020-2021 | Engineering school | ECINE 3-S4<br>International University of Rabat | Electricity | 32 |
| 2020-2021 | Engineering school | ECINE 4-S8<br>International University of Rabat | Renewable Energy Sources | 36 |
| 2020-2021 | Engineering school | ECINE 3-S5<br>International University of Rabat | Automatic: Control & Command | 16 |
| | | | Electronic Signals and Systems: Signal Processing and Conception | 32 |
| 2020-2021 | Engineering school | CPI 2-S4<br>International University of Rabat | Signals and Systems | 40 |
| 2019-2020 | Engineering school | TICLab 2-S3<br>International University of Rabat | Computer Networks | 16 |
| 2019-2020 | Engineering school | ECINE 3-S5<br>International University of Rabat | Automatic: Control & Command | 16 |
| | | | Electronic Signals and Systems: Signal Processing and Conception | 32 |
| 2018-2019 | Engineering school | ECINE 2-S4<br>International University of Rabat | Automatic: Control & Command | 32 |
| 2018-2019 | Bachelor's degree | Master<br>Chouaib Doukkaly University | Computer Networks | 40 |
| 2017-2018 | Bachelor's degree | Bachelor's degree<br>Chouaib Doukkaly University | Telecommunications, Computer Networks | 56 |
| 2016-2017 | Bachelor's degree | Bachelor's degree<br>Chouaib Doukkaly University | Office automation software<br>Literature research<br>Graduation project redaction | 60 |
| **Total** | | | | **911** |

## Co-supervision

- **Postdoctoral:**
- **BOUSSAFA Abdessamad (January2026-present),** "*Power-to-Hydrogen: Optimal Control Strategies for Hydrogen Production/Consumption in a Multi-Source and Multi-Storage Solution Park*", Sciences and Technology Department, UMR Espace-Dev Lab, Energy Transition Team, University of French Guiana, Cayenne, France.
- **PhD student:**

  **SWAIN, Akhyurna (2022-2025),** "*Condition Monitoring and Fault Detection of Offshore Wind Turbine Drive Train*", Department of Electrical and Computer Engineering, New Jersey Institute of Technology, USA.

  **ALIDRISSI Youssef (2020-2024),** "*A Model Predictive Control Strategy for Energy Management in Microgrid System Using Machine Learning Algorithms*". LERMALab, Rabat, International University of Rabat.
- **Undergraduate and graduate students:**
  - ✓ Supervision of master's students in work-study projects in collaboration with companies (France),
  - ✓ Semestrial supervision of undergraduate and graduate students' projects (from 2022 to the present day),
  - ✓ Supervising more than 20 students for graduation projects and mini projects (2017-2022) at the International University of Rabat,
  - ✓ Participating with students in national and international computations about innovation and entrepreneurship.

## Awards & Honors

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| Oct-2022 | Postdoc of the Month: Department of Electrical and Computer Engineering, New Jersey Institute of Technology, United States. |
| 2019 | Best Poster Award: LERMA (Renewable Energies and Materials) Laboratory Seminar held at the International University of Rabat. |
| 2016 | First Class Honors: Master: Industrial Electronics and Telecommunication, Chouaib Doukkali University. |

## Skills

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| Computer & Systems | Hardware and Power Hardware in the Loop (PHIL): OPEL-RT, RTDS, DSpace. |
| | Mastering Simulation tools: MATLAB, PVsyst, RETScreen, EnergyPlus, COMSOL Multiphysics, JMAG-Designer, ISIS Proteus, GNS3, Packet Tracer, and VMWARE WORKSTATION. |
| | Networks Configuration, Architecture, and study of Networks & Routers Configuration. |
| | Programming languages Python, Matlab. |
| | Windows Operating System; Linux. |
| Big-Data/IoT Technologies | Hadoop cluster installation, Node-RED, Thingsboard, Thingspeak. |
| | Data collection and analyses, Real-time data monitoring, and control. |
| Languages | Arabic<br>English : Native.<br>French |

## Links

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| **Google Scholar** | https://scholar.google.fr/citations?user=vPh13ZwAAAAJ&hl=fr |
| **Hal.science** | https://hal.science/search/index?q=abdellatif+elmouatamid |
| **ORCID** | https://orcid.org/my-orcid?orcid=0000-0001-5499-5865 |
| **Scopus** | https://www.scopus.com/authid/detail.uri?authorId=57201975143 |
| **AD Scientific** | https://www.adscientificindex.com/scientist/abdellatif-elmouatamid/4360552 |

## References


**Philip W. T. Pong**
Associate Professor
New Jersey Institute of Technology, United States
philip.pong@njit.edu
Phone: +1 973-596-3533

**Najib Elkamoun**
Associate Professor
Faculty of Sciences, Chouaïb Doukkali University, Morocco
elkamoun.n@ucd.ac.ma
Phone: +212 679332745

**Khalid Zine-Dine**
Associate Professor
Faculty of Sciences, Mohammed V University, Morocco
khalid.zinedine@um5.ac.ma
Phone: +212 661085854